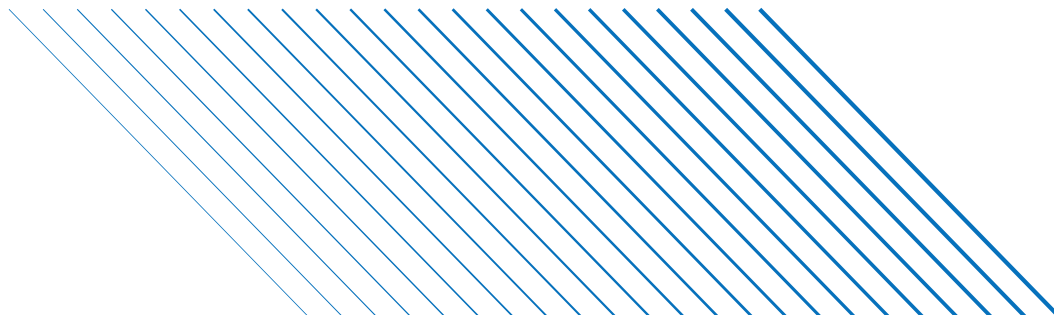
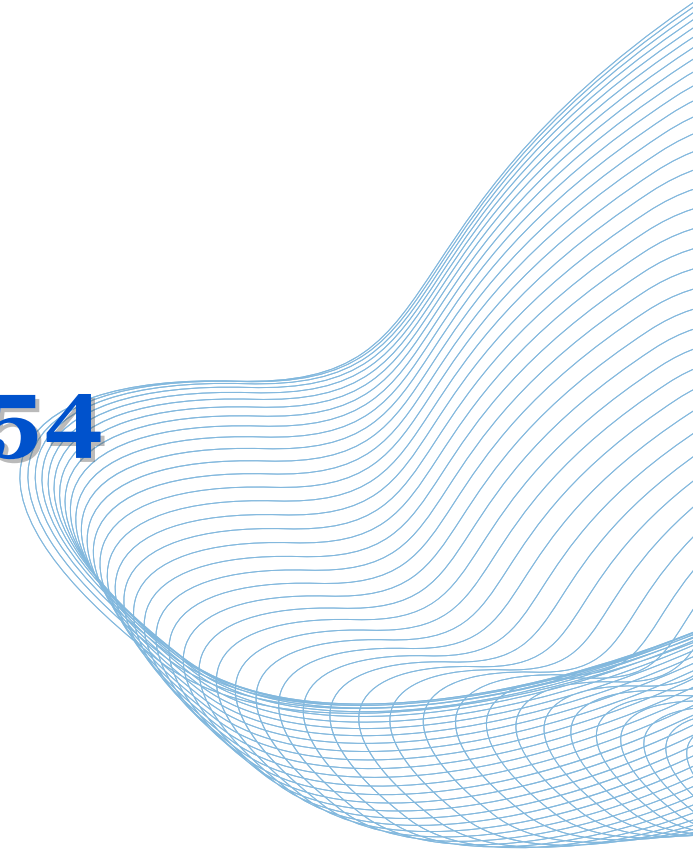


SageArk

Quant Training Session 53-54

The Art And Science of Risk Taking I-II



Intro

- What to expect in this and the next two sessions
- How to be experienced risk takers
- Additional thoughts on risk, not wholesale change

Different Trading Strategies

- Liquidity Provider or Market Maker
- Relative Value
- Prop Trading or Macro

Risks Depend on Strategy

- Shorter holding period is more dependent on liquidity
- Longer holding period means more dependent on edge identification and time and path to realize edge

Why would you buy or sell?

- The price you can trade is different from your idea of 'fair value'
 - $\text{Edge} = \text{Mkt Price} - \text{Fair Value}$
- What is 'fair value'
 - Market mid? Join the bid offer
 - Assume relationship like Cash vs. Futures basics etc.
 - Fundamental difference with today's price based on research, model, or other framework for value

Realizing Value

- Buy and then another person buys from you
- Buy and manage position until some relationship moves to your target
- Buy and then exit over time as facts change
- Difference in these is holding period, outcome
 - More time, more variance
 - More variance = PNL swings
 - OPM?

Easy?

- How do you know for sure?
- How confident are you in your model of the prices?
- How confident are you in your estimation of all the possible outcomes and their probabilities?
- What events would change your estimate of fair value?

Intro

- Keep your seat
- Communication and Trust
- Expected value is not enough to evaluate a trade
- 2 coins under my hand example

Risk

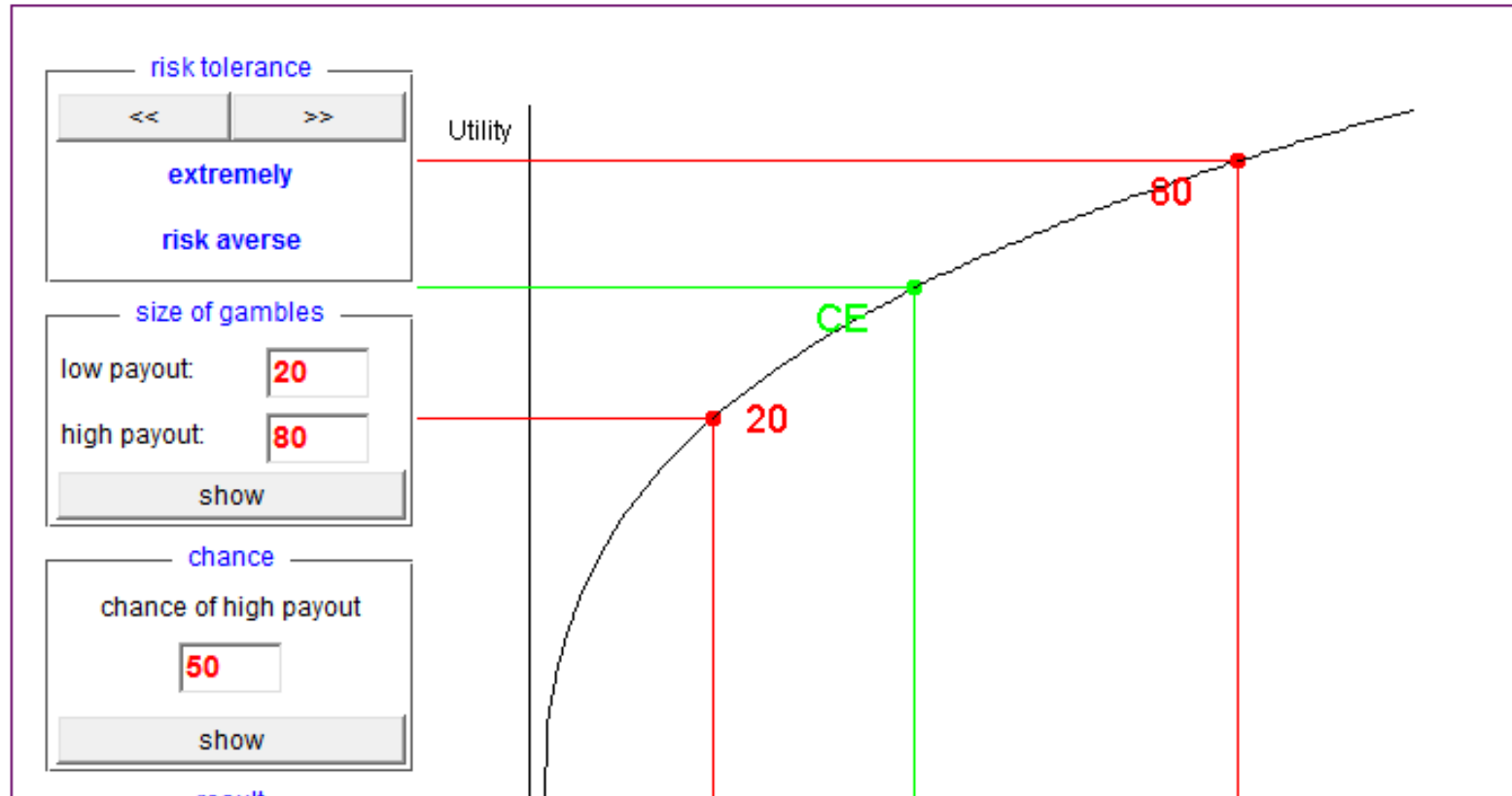
- Do we care only about long term wealth accumulation?
 - If yes, then Kelly applies
 - If there are constraints on the path along which we walk toward maximizing wealth, then what?

Utility of Wealth

- St. Petersburg Paradox
- Creation of Utility Theory
 - The last \$1MM is not as sweet as the first
 - $U(w) = \text{Log}(w)$
- Prospect theory and risk aversion

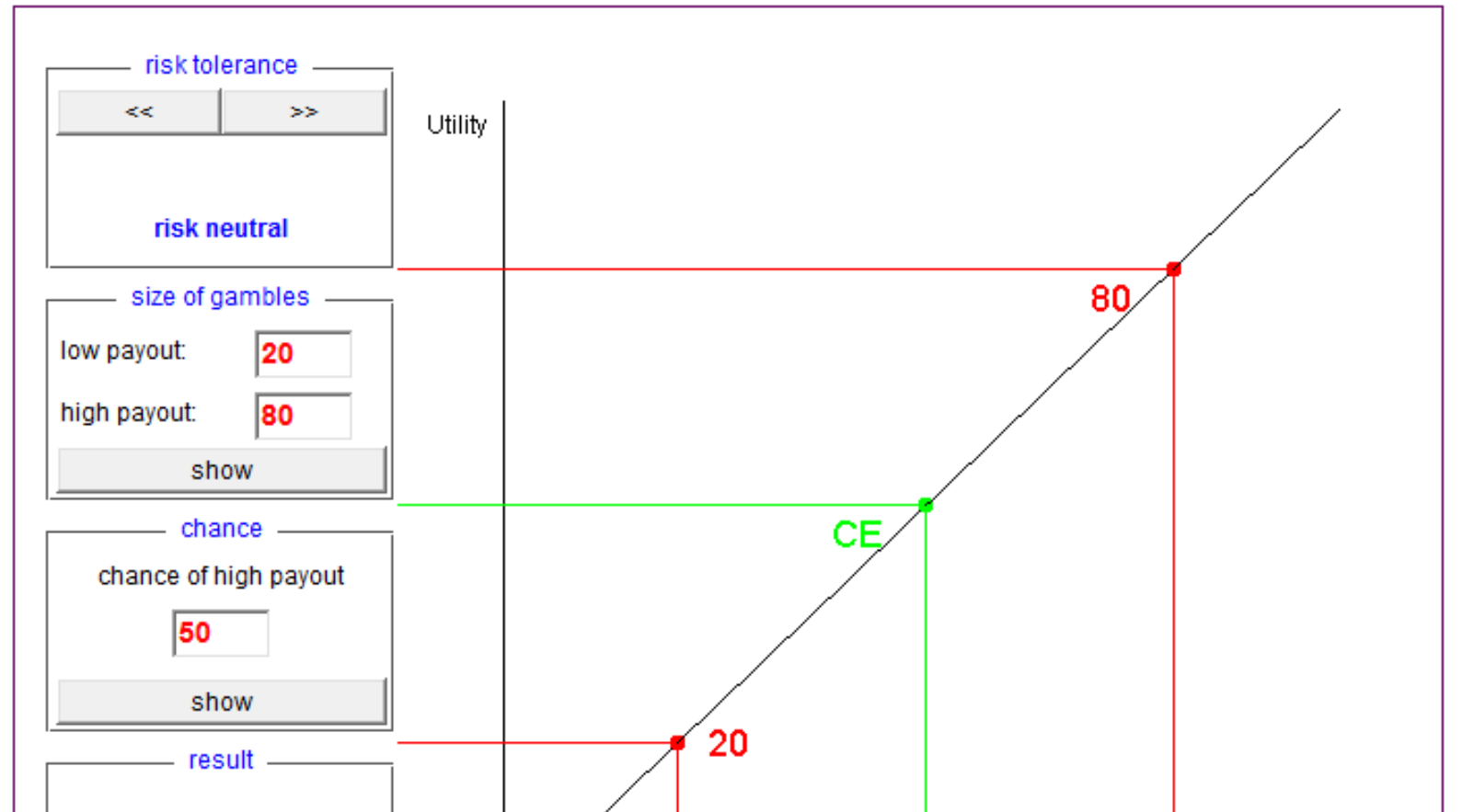
Moderate Risk Aversion

Risk and Certainty Equivalence Applet
(see instructions)



No Risk Aversion

Risk and Certainty Equivalence Applet (see instructions)



When Should I Slow Down or Stop?

- When is it still credible that I have 'edge'?
 - When can we reject the hypothesis?
- Have a plan and stick to it
 - Many people say, "Don't add to losers"
 - Trade by trade set a time or framework, so you can know when it's not working

How Much Should I Lose?

- Rules of Thumb – Coping mechanisms
 - Half of what you aim to make
 - 2 months income/returns
 - No more than 20% of your total stack in one
 - What outcome makes me sad
 - Lose your job
 - Not having enough \$
 - Not achieving max wealth

Trade Sizing

- Trade sizing should be relative to your edge
- Trade sizing should be relative to the risk
- Trade sizing should be relative to your “stop”
 - Where here “stop” can mean different things

Kelly Criterion

- Kelly Criterion as a way to estimate trade sizing
- Additional reading

Trust

- Can I explain *ahead of time* how risky my trades will be?
- Can I control the risk as things inevitably shift around?
- Control Risk = Limit the Loss

Winner's Game vs. Loser's Game

- Ellis article
- Tennis
 - Pros win points, Amateurs lose points
- Golf
 - Win by making fewer bad shots (loser's game)
- Football
 - Score more points? Give up fewer?
- Trading
 - Don't give up any points
 - Keep it simple and do things where you have an advantage

Risk

- Dealing with the future
- Financial theory says Risk = Volatility
- Most of our discussions will detail why this is at best, inadequate

Risk

- We only ever experience one outcome of the many that were possible
- Unlike dice or cars, we never ‘know’ what else might have happened.
- How can we measure risk?
 - We often get anchored on the near future being like the near past

Risk

- Rarely hear someone say
 - I will not buy this as it might show large fluctuations
- More sensibly
 - I need more upside potential because I am afraid I might lose money
- The main risk is that of losing money
 - Let's not overcomplicate things (yet)

Betting, Trading, Investing

- Bet on a horse – It's gambling
 - Bet on a flush draw – It's entertainment
 - Short Nat Gas Oct \$3.00 puts – It's trading
 - Long the SPX – It's investing
-
- Closed vs. Open systems
 - Lots of analogies, but...
 - Know what you know. Know what you don't know.

Risk

- Always about the future
 - More things can happen than actually will happen
 - Only realize one value of many possibilities
- Risk is invisible in a way, covert, a possibility of loss. Subjective.
- Losses are real, observable, and the product of positions crossing with adverse market moves

Probably Not

- Probable things fail to happen all the time
 - 70% chance of rain
 - Long shots at track
- Improbable things actually do happen
- These things usually happen for two reasons...
 - Unlucky 'roll of the dice'
 - Our estimate or understanding of the odds was 'off'

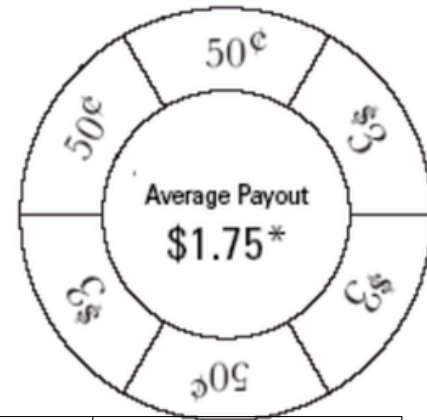
Risk

- Performance is not a great way to measure risk
- Bad performance indicates bad outcome
 - Could be for many reasons
- Good performance says little about the risks taken and not realized

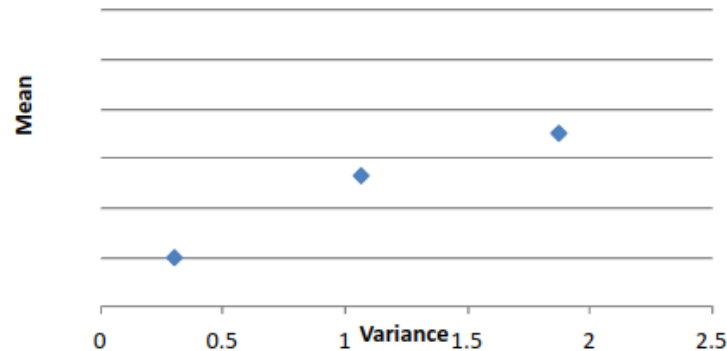
Risk

- The past can give you a guide...
 - There will always be some uncertainty
 - Trading is not a closed system
 - Things change. Often more than you think.
- Things do happen that never happened before
 - LTCM (bond / old bond + 30bp)
 - National Housing price decline
 - Two 40% Equity Index declines in 10 years

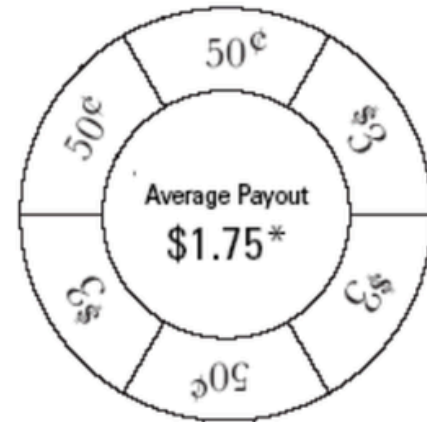
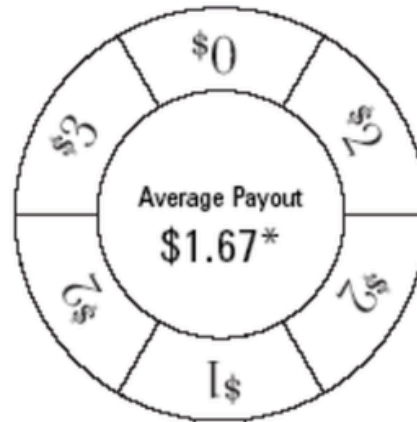
Let's Do Some Investing



Variance	0.3		1.066667		1.875
----------	-----	--	----------	--	-------



Let's Do Some Investing



Geometric Mean	1.414214	0	1.224745
----------------	----------	---	----------

Run this investment thru 52 'trials' or 'weeks' and see which one you like better...where you let your profits ride.

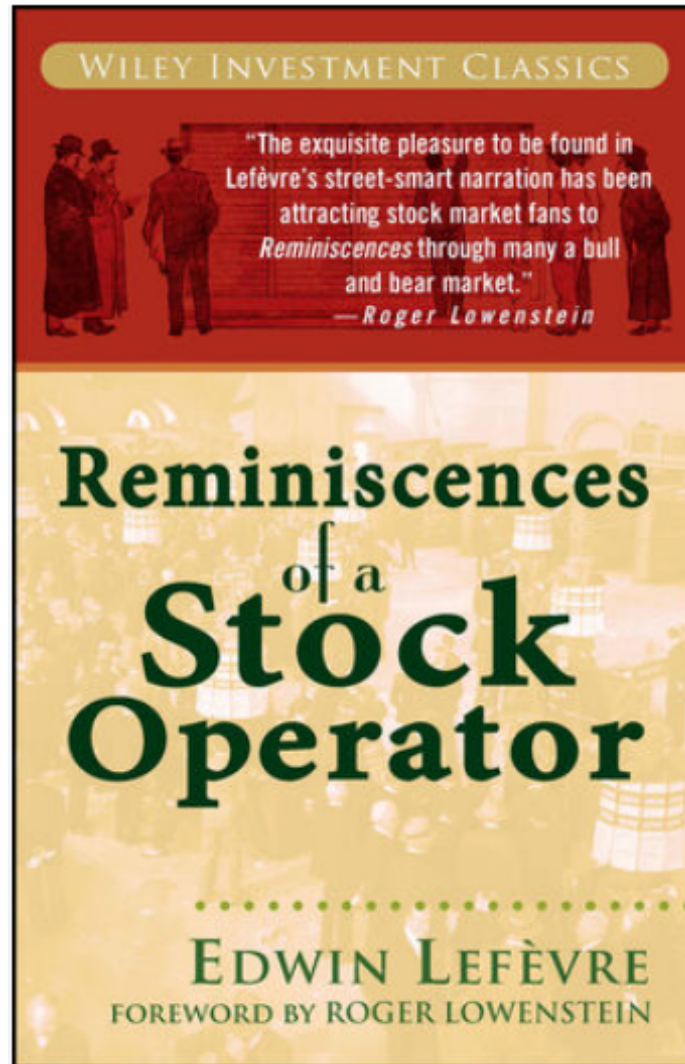
Risk

- Keep Variance in check is not enough to keep you from losing too much money
- Are you going to play more than once?
 - More is better if you have positive expectation
 - More outcomes mean more likely to realize expected value



Decision Making & Pyschology

Book Recommendation



Eternal Importance of Trading Psychology

It seems incredible that knowing the game as well as I did and with an experience of twelve or fourteen years of speculating in stocks and commodities I did precisely the wrong thing.

The cotton showed me a loss and I kept it. The wheat showed me a profit and I sold it out. It was an utterly foolish play.... Always sell what shows you a loss and keep what shows you a profit.

That was so obviously the wise thing to do and was so well known to me that even now I marvel at myself for doing the reverse.

Class Outline

- Eternal importance of trading psychology
- Doing your own work = The only way to have true conviction
- Getting personal/emotional with the market + Being emotionally mature
- Danger of overtrading
- Don't understand? De-risk
- Hope is not a strategy
- Bias trading
- Appreciating the bigger picture
- Other classic mistakes

Doing Your Own Work = The Only Way to Have True Conviction

- Speculation is a hard and trying business, and a speculator must be on the job all the time or he'll soon have no job to be on.
- A man cannot be convinced against his own convictions, but he can be talked into a state of uncertainty and indecision, which is even worse, for that means that he cannot trade with confidence and comfort.

Getting Personal/Emotional with the Market + Being Emotionally Mature

- I never argue with the tape. To be angry at the market because it unexpectedly or even illogically goes against you is like getting mad at your lungs because you have pneumonia.
- Losing money is the least of my troubles. A loss never bothers me after I take it. I forget it overnight. But being wrong -- not taking the loss that is what does the damage to the pocketbook and to the soul.

Danger of Overtrading

I always made money when I was sure I was right before I began. **What beat me was not having brains enough to stick to my own game -- that is, to play the market only when I was satisfied that precedents favored my play.**

There is a time for all things, but I didn't know it. And that is precisely what beats so many men in Wall Street who are very far from being in the main sucker class. There is the plain fool, who does the wrong thing at all times everywhere, but there is the Wall Street fool, who thinks he must trade all the time.

No man can always have adequate reasons for buying or selling stocks daily or sufficient knowledge to make his play an intelligent play.

Don't Understand? De-risk

- It is enough for the experienced trader to perceive that something is wrong. He must not expect the tape to become a lecturer. His job is to listen for it to say "Get out!" and not wait for it to submit a legal brief for approval.

Hope is not a Trading Strategy

The speculator's chief enemies are always boring from within. It is inseparable from human nature to hope and to fear. **In speculation when the market goes against you -- you hope that every day will be the last day and you lose more than you should had you not listened to hope -- to the same ally that is so potent a success-bringer to empire builders and pioneers, big and little.**

And when the market goes your way you become fearful that the next day will take away your profit, and you get out too soon. Fear keeps you from making as much money as you ought to. The successful trader has to fight these two deep-seated instincts. He has to reverse what you might call his natural impulses. Instead of hoping he must fear; instead of fearing he must hope. He must fear that his loss may develop into a much bigger loss, and hope that his profit may become a big profit.

Bias Trading

- It takes a man a long time to learn all the lessons of all his mistakes. They say there are two sides to everything. But there is only one side to the stock market; and it is not the bull side or the bear side, but the right side.
- I was utterly free from speculative prejudices. The bear side doesn't appeal to me any more than the bull side, or vice versa. My one steadfast prejudice is against being wrong.

Appreciating the Bigger Picture

- One of the most helpful things that anybody can learn is to give up trying to catch the last eighth or the first. These two are the most expensive eighths in the world. They have cost stock traders, in the aggregate, enough millions of dollars to build a concrete highway across the continent.
- Old Baron Rothschild's recipe for wealth winning applies with greater force than ever to speculation. Somebody asked him if making money in the Bourse was not a very difficult matter, and he replied that, on the contrary, he thought it was very easy....“I will tell you my secret if you wish. It is this: I never buy at the bottom and I always sell too soon.”

Other Classic Mistakes

- Letting the PNL asymmetry affect you
- Trading to marks
- Trading to prices
- Fighting the trend
- Not recognizing regime change
- Focusing too narrowly on your own market
- <Insert any one of a million other potential mistakes>

Long Term Capital Management

Long Term Capital Management

- Models...
- When Genius Failed
 - They were in-fact geniuses
 - They had developed models which were tested and produced large profits over several years
 - What could go wrong?

Long Term Capital Management

- Perceptions change – often rapidly
- What are the limits or boundaries for prices or relationships?
 - What do you know for sure?
- What are other people doing?
- Are you too big?
- Where did all the liquidity go?

Bond vs. Old Bond



Fed Buying Mortgages



Fed Buying Mortgages



Crude Outright Price



Definitely outside of expectations of almost everyone. Why?



EDM5



Things Change...

And we aren't always good at anticipating changes

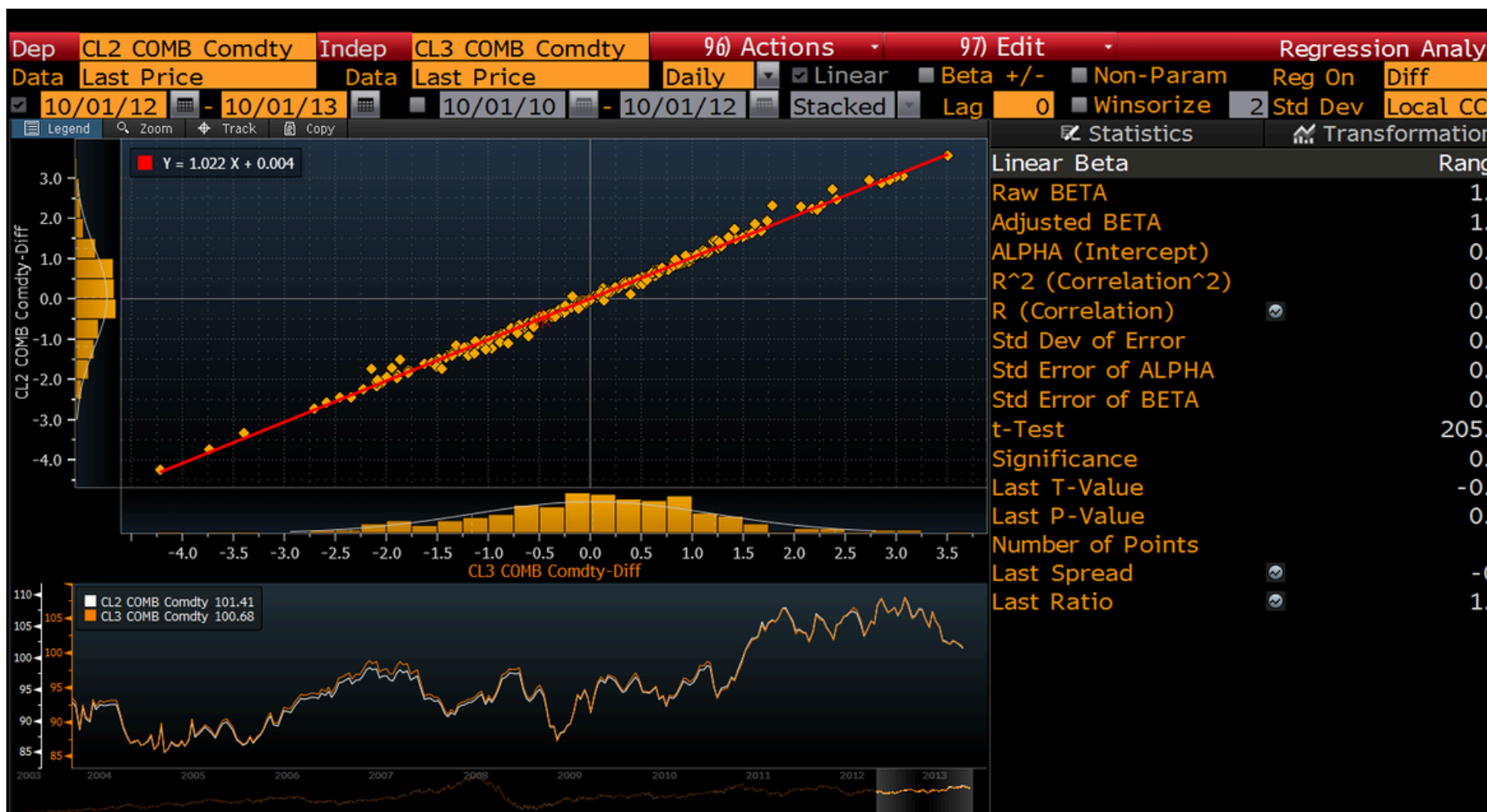


Other Portfolio/Trading Accidents

- LTCM
- DE Shaw
- Bear Stearns
- Amaranth
- Fannie Mae
- JP Morgan (Whale)

- WHY does this happen all the time??
 - Discuss

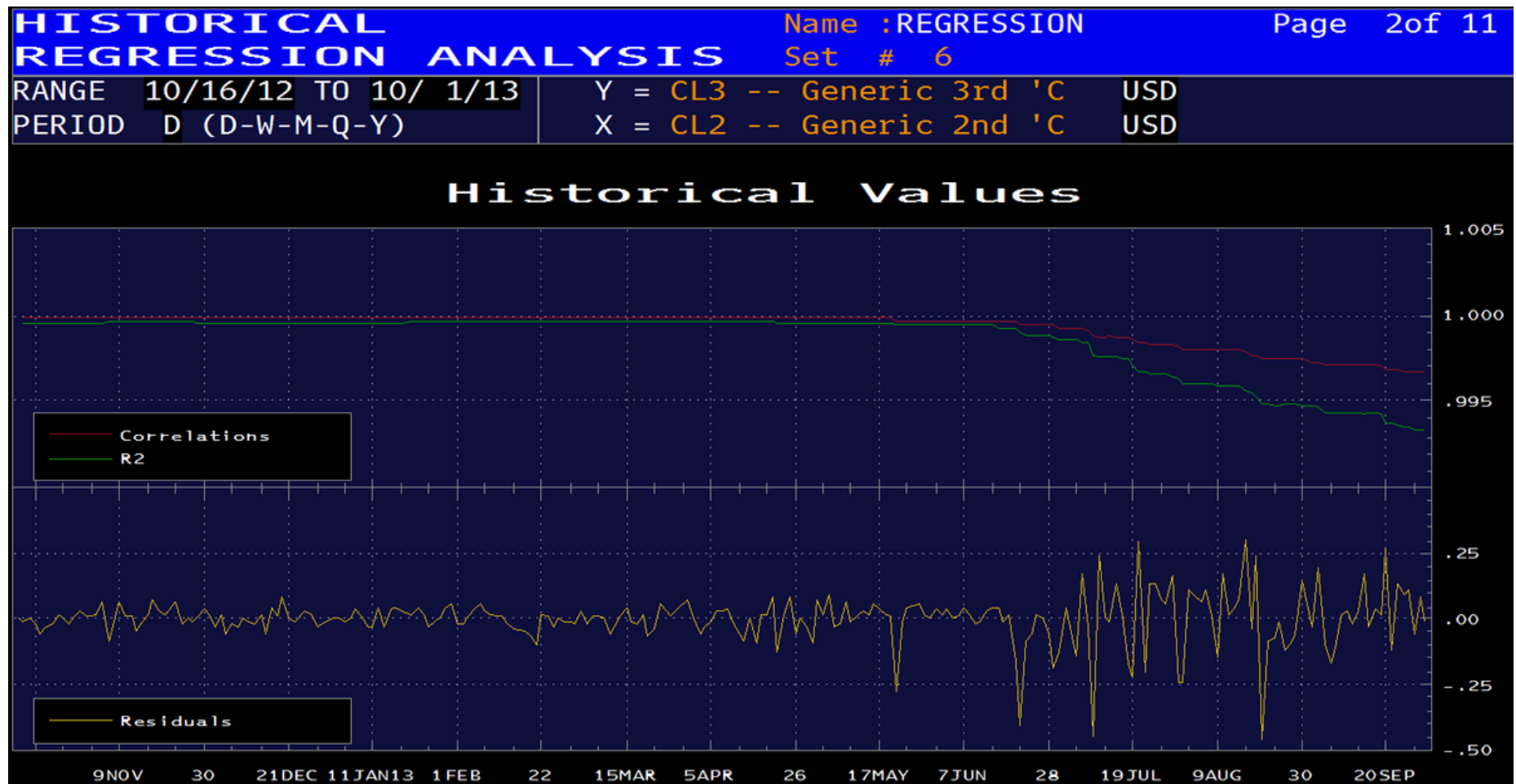
Appendix: How far can it move?



Appendix: The Spread



Appendix: The residuals, r^2 etc



Appendix

Upside	Downside	P[Upside]	Horizon Yr's	Mean	Variance	Risk	Ann. Sharpe	Kelly Criteria bet size	CE
<i>Symmetric return profile with varying conviction levels</i>									
20%	-20%	40%	0.25	-4.00%	3.8%	20%	(0.41)	-20.0%	-6%
20%	-20%	50%	0.25	0.00%	4.0%	20%	0.00	0.0%	-2%
20%	-20%	60%	0.25	4.00%	3.8%	20%	0.41	20.0%	2%
20%	-20%	70%	0.25	8.00%	3.4%	18%	0.87	40.0%	6%
20%	-20%	80%	0.25	12.00%	2.6%	16%	1.50	60.0%	11%
20%	-20%	90%	0.25	16.00%	1.4%	12%	2.67	80.0%	15%

Appendix

Upside	Downside	P[Upside]	Horizon Yr's	Mean	Variance	Risk	Ann. Sharpe	Kelly Criteria bet size	CE
Significant upside asymmetry in the payoff profile, calibrated to earn 5%									
20%	-10%	50%	0.50	5.00%	2.3%	15%	0.47	25.0%	4%
30%	-10%	38%	0.50	5.00%	3.8%	19%	0.37	16.7%	3%
40%	-10%	30%	0.50	5.00%	5.3%	23%	0.31	12.5%	2%
50%	-10%	25%	0.50	5.00%	6.8%	26%	0.27	10.0%	2%
60%	-10%	21%	0.50	4.98%	8.2%	29%	0.25	8.3%	1%
70%	-10%	19%	0.50	5.04%	9.8%	31%	0.23	7.2%	0%

Appendix: Example of Bayes' Theorem

(Or How to Update My Fair Value Model)

	Input Data	User Fills in the Red Cells			
	Prior Probability Forecast				
	Mean of my Forecast	170			
	Time Horizon (weeks)	8			
	50% Range of my Forecast (+or -)	7			
	Price Path				
	Gain/(Loss) per Week	0.63			
	Forecast - Inception	5			
					Decision Point:
Week #	Prices Over Time	Actual Prices	Assumed Path	Actual-Assumed	Mean of Distribution
0	At Inception	165	165.00	0.00	170.00
1	After 1 Week	164.13	165.63	-1.50	168.94
2	After 2 Week	160.33	166.25	-5.92	166.43
3	After 3 Week	162.02	166.88	-4.85	164.72
4	After 4 Week	160.56	167.50	-6.94	162.89

Sorry, we couldn't load this chart.

Appendix: Risk and Compounding

- Compounding is really and truly the key
 - The larger the drawdown, the more work you have to do to make it back
 - -50%, then up 50%...means you are still down 25%***
 - Volatility is costly due to this
 - +1%, -1% daily for a year. Are you flat PNL?
 - See appendix

Appendix: Positive and Negative Returns

252 Days

		sum of ups and downs		
	100	0		
1	99	-0.01	-1	
2	99.99	0.01		
3	98.9901	-0.01		
4	99.98	0.01		
5	98.9802	-0.01		
6	99.97	0.01		
7	98.9703	-0.01		
8	99.96001	0.01		
9	98.96041	-0.01		
10	99.95001	0.01		
11	98.95051	-0.01		
248	98.7676	0.01		
249	97.77992	-0.01		
250	98.75772	0.01		
251	97.77014	-0.01		
252	98.74784	0.01		